

First Registrars & Investor Services Limited

API Integration & Systems Architecture Documentation

Executive Summary

This document provides a technical overview of the existing API surface, third-party integrations, infrastructure dependencies, and system communication architecture used within the [First Registrars & Investor Services Limited digital ecosystem](#).

While the current platform does not expose a formal Swagger/OpenAPI specification, the application contains multiple operational endpoints, integration services, external APIs, and supporting infrastructure components that collectively power shareholder services, payment workflows, communication systems, and investor-facing digital products.

This document serves as a strategic technical reference for:

- technology partners,
 - integration teams,
 - infrastructure architects,
 - software vendors,
 - and enterprise stakeholders.
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1. System Overview

Platform Architecture

The platform is primarily implemented as a Microsoft ASP.NET Core MVC web application running on .NET 8.

The ecosystem integrates multiple supporting systems including:

- SQL Server
- MongoDB
- SMTP Email Infrastructure
- Payment Gateways
- Shareholder Access Portals

- Live Chat Services
 - External APIs
 - Document Delivery Systems
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2. Core Technology Stack

Component	Technology
Backend Framework	ASP.NET Core MVC (.NET 8)
Programming Language	C#
ORM	Entity Framework Core
Primary Database	Microsoft SQL Server
Secondary Database	MongoDB
Frontend Rendering	Razor Views
Payment Gateway	Paystack
Email Infrastructure	Microsoft 365 SMTP
Live Chat Integration	Ebanqo
Hosting Model	Azure-Compatible Architecture

3. Existing API Surface

Overview

The current application does not include:

- Swagger UI,
- OpenAPI documentation,
- or a standalone REST API specification.

Instead, the operational API surface is derived from:

- MVC controller routes,
 - integration endpoints,
 - document delivery endpoints,
 - and service interaction layers.
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4. Website Route Endpoints

The primary public endpoints are exposed through `HomeController.cs`.

4.1 Corporate Information Endpoints

Method	Endpoint	Description
GET	/	Home page
GET	/about	About page
GET	/about/team	Management team
GET	/about/team/{id}	Team member details
GET	/about/directors	Directors listing
GET	/about/director/{id}	Director details
GET	/about/management-team	Management team listing
GET	/about/management-team/{id}	Management detail
GET	/about/clients	Client listing

4.2 Service Endpoints

Method	Endpoint	Description
GET	/services	Services overview
GET	/services/register-management	Registrar services
GET	/services/kyc-verification	KYC verification
GET	/services/electronic-voting	Electronic voting
GET	/services/probate	Probate services

4.3 e-Product Endpoints

Method	Endpoint	Description
GET	/e-products	Products overview
GET	/products	Alias route
GET	/e-products/online-access	Online access
GET	/e-products/e-share-notifier	Share notifier
GET	/e-products/e-dividend	E-dividend
GET	/e-products/first-dividend-plus-card	Dividend card
GET	/e-products/m-access	M-access
GET	/e-products/mobile	Mobile app
GET	/e-products/e-lodgement	E-lodgement

4.4 Resource & Media Endpoints

Method	Endpoint	Description
GET	/blog/{cat?}	Blog listing
GET	/post/{id}	Blog detail
GET	/annual-reports	Reports listing
GET	/download-report/{filename}	Report download
GET	/faq	FAQ page
GET	/media	Media gallery
GET	/forms	Forms page

4.5 Contact & Policy Endpoints

Method	Endpoint	Description
GET	/contact	Contact page
GET	/privacy	Privacy policy
GET	/whistle-blowing-policy	Whistle blowing
GET	/terms	Terms & conditions
GET	/online-access	Redirect to portal
GET	/subscribe	Newsletter subscription
GET	/error	Error fallback

5. Core System Integrations

5.1 SQL Server & Entity Framework Core

Overview

The primary application database is powered by Microsoft SQL Server using Entity Framework Core.

Key Characteristics

Item	Value
ORM	Entity Framework Core

Item	Value
Database	SQL Server
DB Context	AppDB
Configuration Key	DefaultConnection

Responsibilities

The SQL Server database is responsible for:

- core application data,
- user records,
- content management,
- transactional workflows,
- reporting,
- and platform operations.

Relevant Components

- `Startup.cs`
- `_db.cs`

5.2 MongoDB Integration

Overview

MongoDB support exists within the core library for supplemental and scalable document-oriented storage requirements.

MongoDB Components

Item	Value
Client Wrapper	Mongo
Driver	IMongoClient
Database Name	frdb
Config Key	mongouri

Potential Use Cases

- logs,
- semi-structured records,
- activity tracking,
- caching,
- or auxiliary data storage.

Relevant Components

- `_mongo.cs`
 - `appsettings.json`
 - `appsettings.Production.json`
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5.3 Email Delivery Infrastructure

Overview

The platform uses Microsoft 365 SMTP infrastructure through FluentEmail for transactional and operational communications.

SMTP Configuration

Item	Value
SMTP Host	smtp.office365.com
SMTP Port	587
SSL	Enabled
Sender Account	friscomms@firstregistrarsnigeria.com

Supported Email Operations

- account confirmation,
- welcome emails,
- password reset emails,
- payment notifications,
- ticket communications,
- share offer notifications.

Relevant Component

- `_mail.cs`
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5.4 Paystack Payment Integration

Overview

The platform integrates with [Paystack](#) for payment verification and transaction processing.

Integration Components

Component	Purpose
Paystack Inline JS	Payment collection
Verification API	Transaction verification
Core Parser	Payment response parsing

External Endpoints

URL	Purpose
https://js.paystack.co/v1/inline.js	Inline payment script
https://api.paystack.co/transaction/verify	Payment verification

Relevant Components

- `index.cs`
- `paystack.cs`

5.5 Shareholder Access Portal Integration

Overview

The website integrates with an external shareholder access platform.

Configuration

Item	Value
Config Key	<code>accessurl</code>
Production URL	https://access.firstregistrarsnigeria.com
Website Endpoint	<code>/online-access</code>

Functionality

This integration enables:

- shareholder login,
- investor account access,
- digital account services,
- and self-service shareholder workflows.

Relevant Component

- `HomeController.cs`

5.6 Ebanqo Live Chat Integration

Overview

The platform integrates with [Ebanqo](#) for customer communication and live support.

Integration URLs

URL	Purpose
<code>https://widget.ebanqo.io/app.js</code>	Chat widget loader
<code>https://webchat.ebanqo.io/firstregistrars</code>	Hosted webchat

Functionality

The widget enables:

- customer support,
- real-time inquiries,
- investor assistance,
- and service communication.

Relevant Component

- `_Layout.cshtml`

5.7 Report & File Delivery System

Overview

The platform includes a file delivery subsystem for reports and downloadable documents.

Delivery Features

Feature	Description
Annual Report Downloads	Public investor reports
Blob-style Path Resolution	Dynamic file retrieval

Feature	Description
Download Routing	File delivery endpoints

Key Helpers

- `Tools.ReportPath`
- `GetDownloadPath(...)`
- `GetBlobFilePath(...)`

Relevant Components

- `HomeController.cs`
- `index.cs`

6. External Web Integrations

The ecosystem currently references several external systems and digital services.

Active Integrations

System	Purpose
<code>firstregistrarsnigeria.com</code>	Main website
<code>access.firstregistrarsnigeria.com</code>	Shareholder portal
<code>fr-access-api.azurewebsites.net</code>	API services
<code>docs.firstregistrarsapi.com</code>	API documentation portal
<code>check.cle.ng</code>	Verification/check services
<code>cle.ng</code>	Supporting ecosystem
Facebook	Social engagement
Instagram	Brand presence
X/Twitter	Public communication
Cookie Consent JS	Privacy compliance

7. Configuration Architecture

Key Configuration Values

The application uses centralized configuration management through application settings.

Key	Purpose
DefaultConnection	SQL Server connection
mongouri	MongoDB connection
accessurl	Shareholder portal URL
apiurl	External API base URL

Relevant Files

- appsettings.json
- appsettings.Production.json
- fn.cs

8. Architectural Observations

Strengths of the Current Integration Model

Multi-System Ready

The platform already supports multiple enterprise integrations and can scale into a broader digital registrar ecosystem.

Hybrid Data Architecture

Combining SQL Server and MongoDB creates flexibility for both structured and semi-structured workloads.

Enterprise Communication Infrastructure

Microsoft 365 SMTP provides reliable enterprise-grade email delivery.

Payment Ecosystem Readiness

Paystack integration enables scalable digital financial workflows.

Modular Expansion Potential

The architecture supports future:

- REST APIs,
- mobile apps,
- microservices,
- and external partner integrations.

9. Current Limitations

Missing API Governance Components

The current implementation does not yet expose:

- Swagger/OpenAPI documentation,
- centralized API gateway management,
- formal API versioning,
- API authentication standards documentation,
- or developer onboarding documentation.

10. Recommended Future Enhancements

Short-Term Recommendations

Swagger/OpenAPI Implementation

Introduce formal API documentation using Swagger/OpenAPI.

API Authentication Standardization

Implement:

- JWT authentication,
- OAuth2,
- API keys,
- or Azure AD integration.

Logging & Monitoring

Add:

- centralized logging,
- telemetry,
- audit monitoring,
- and health diagnostics.

Medium-Term Recommendations

API Gateway Layer

Implement:

- Azure API Management,
- request throttling,
- caching,
- and analytics.

Service Segmentation

Separate services into:

- authentication services,
- payment services,
- communication services,
- and investor services.

Cloud Optimization

Introduce:

- Azure Blob Storage,
- CDN optimization,
- serverless workflows,
- and automated scaling.

Long-Term Digital Transformation

Full API-Driven Ecosystem

Transition toward:

- mobile-first architecture,
- headless frontend support,
- partner integrations,
- and fintech interoperability.

Enterprise Integration Platform

Support:

- banking APIs,
 - registrar interoperability,
 - shareholder automation,
 - and real-time investor services.
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11. Strategic Technology Positioning

The current ecosystem already demonstrates a strong enterprise-ready foundation capable of supporting:

- digital registrar modernization,
- investor self-service platforms,
- payment automation,
- communication workflows,
- and future fintech integrations.

The architecture provides an excellent base for evolving into a fully API-driven registrar and shareholder management ecosystem.

12. Conclusion

The First Registrars integration architecture combines:

- modern Microsoft technologies,
- enterprise communication systems,
- payment integrations,
- database flexibility,
- and scalable infrastructure patterns.

Although formal API governance is not yet fully implemented, the existing platform architecture already supports significant operational capability and future enterprise expansion.

With additional modernization initiatives, the ecosystem can evolve into a comprehensive digital registrar platform with:

- secure APIs,
 - partner integrations,
 - investor automation,
 - and cloud-native scalability.
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13. Technical Reference Summary

Component	Technology
Backend Framework	ASP.NET Core MVC
Primary Database	SQL Server
Secondary Database	MongoDB

Component	Technology
Email Infrastructure	Microsoft 365 SMTP
Payment Gateway	Paystack
Live Chat	Ebanqo
Hosting Compatibility	Azure
Authentication Readiness	Extendable
API Readiness	Partial
Architecture Style	Enterprise MVC